

10.1 DC: Practical Circuits & Kirchhoff's Laws

Question Paper

Course	CIEA Level Physics
Section	10. D.C. Circuits
Topic	10.1 DC: Practical Circuits & Kirchhoff's Laws
Difficulty	Medium

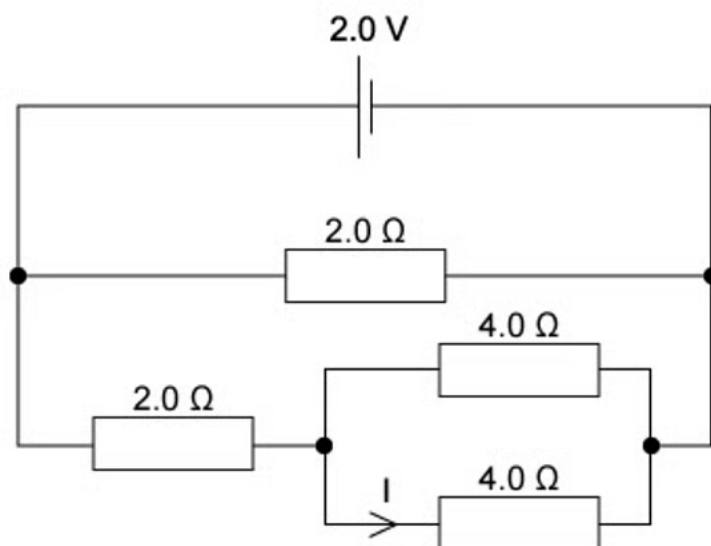
Time allowed: 20

Score: /10

Percentage: /100

Question 1

A cell of e.m.f. 2.0 V and negligible internal resistance is connected to a network of resistors as shown.



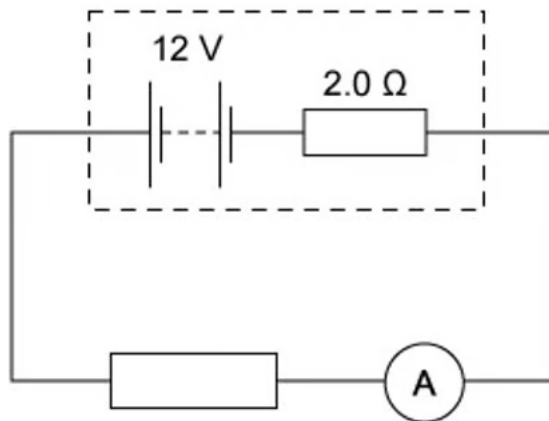
What is the current I ?

- A. 0.25 A
- B. 0.33 A
- C. 0.50 A
- D. 1.5 A

[1 mark]

Question 2

A battery of e.m.f. 12 V and internal resistance $2.0\ \Omega$ is connected in series with an ammeter of negligible resistance and an external resistor. External resistors of various different values are used.



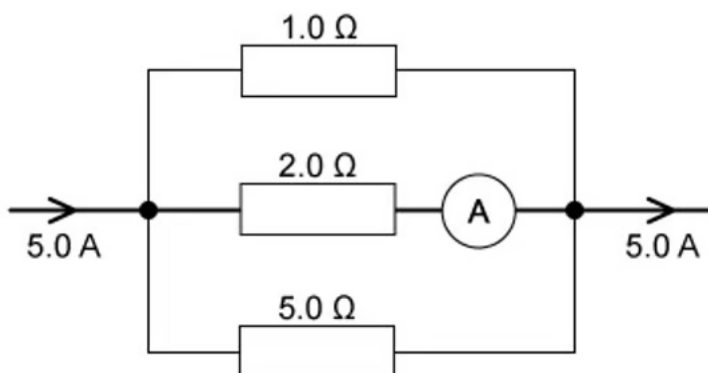
Which combination of current and resistor value is **not** correct?

	Current / A	External resistor value / Ω
A	1.0	10.0
B	1.2	8.0
C	1.5	6.0
D	1.8	4.0

[1 mark]

Question 3

The diagram shows part of a current-carrying circuit. The ammeter has negligible resistance.



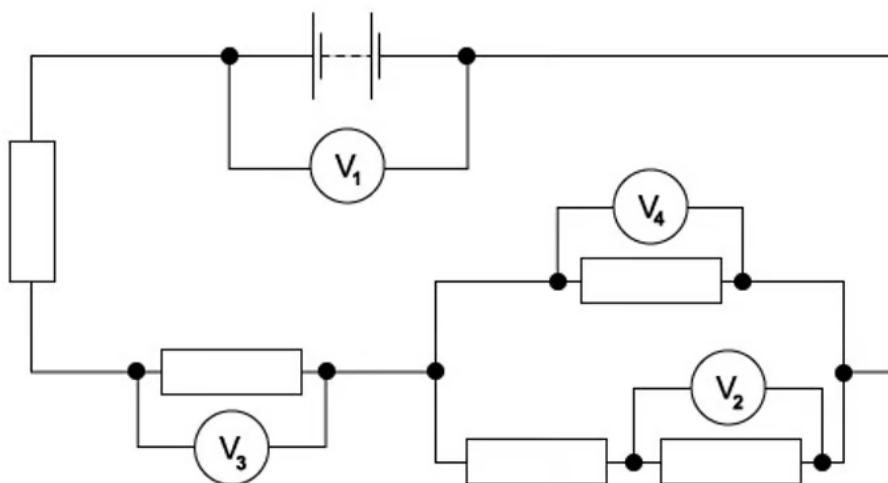
What is the reading on the ammeter?

- A. 0.7 A
- B. 1.3 A
- C. 1.5 A
- D. 1.7 A

[1 mark]

Question 4

In the circuit shown, all the resistors are identical.



The reading on voltmeter V_1 is 8.0 V and the reading on voltmeter V_2 is 1.0 V. What are the readings on the other voltmeters?

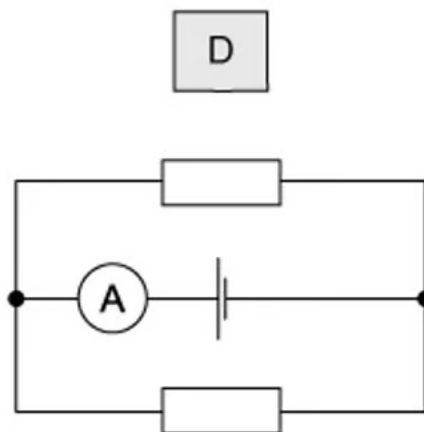
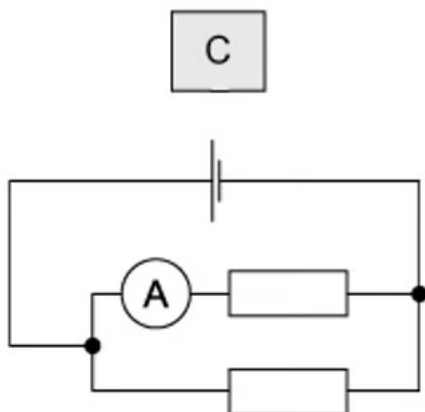
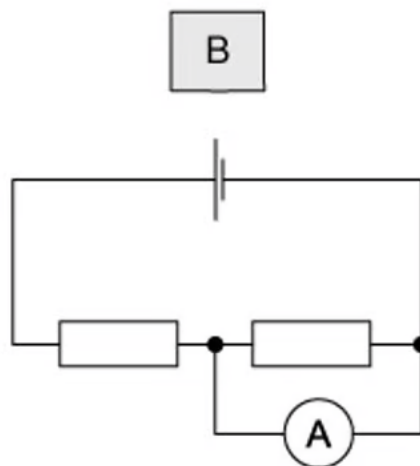
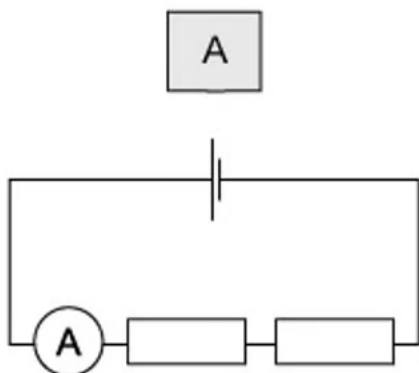
	Reading on voltmeter V_3 / V	Reading on voltmeter V_4 / V
A	1.5	1.0
B	3.0	2.0
C	4.5	3.0
D	6.0	4.0

[1 mark]

Question 5

A cell, two resistors of equal resistance and an ammeter are used to construct four circuits. The resistors are the only parts of the circuits that have resistance.

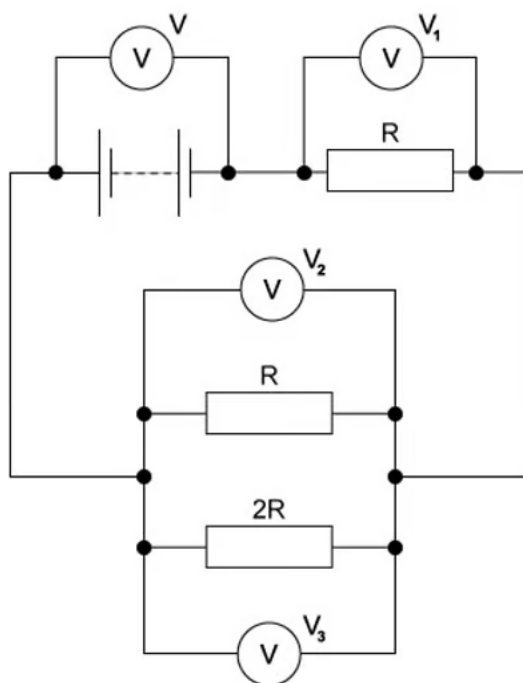
In which circuit will the ammeter show the greatest reading?



[1 mark]

Question 6

The diagram shows a circuit with four voltmeter readings V , V_1 , V_2 and V_3 .



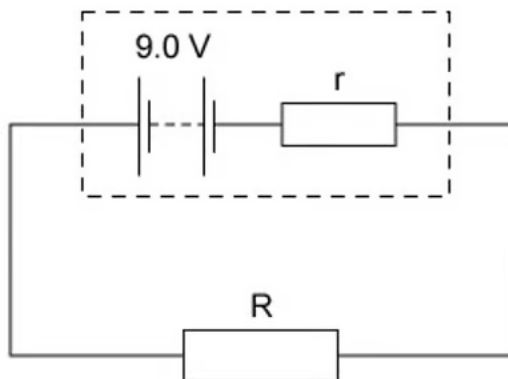
Which equation relating the voltmeter readings must be true?

- A. $V = V_1 + V_2 + V_3$
- B. $V + V_1 = V_2 + V_3$
- C. $V_3 = 2(V_2)$
- D. $V - V_1 = V_3$

[1 mark]

Question 7

A simple circuit is formed by connecting a resistor of resistance R between the terminals of a battery of electromotive force (e.m.f.) 9.0 V and constant internal resistance r .



A charge of 6.0 C flows through the resistor in a time of 2.0 minutes, causing it to dissipate 48 J of thermal energy.

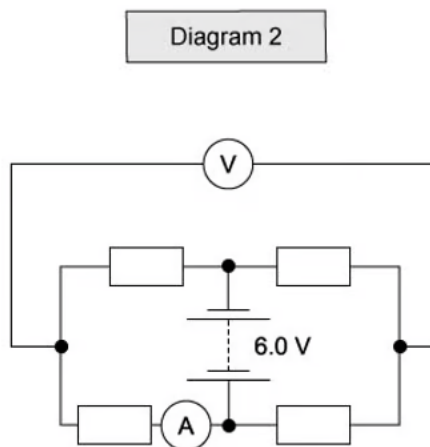
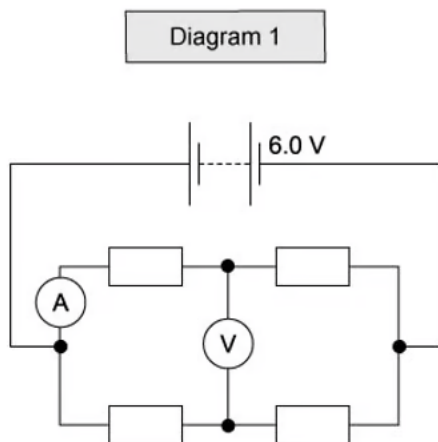
What is the internal resistance r of the battery?

- A. $0.17\ \Omega$
- B. $0.33\ \Omega$
- C. $20\ \Omega$
- D. $160\ \Omega$

[1 mark]

Question 8

When four identical resistors are connected as shown in diagram 1, the ammeter reads 1.0 A, and the voltmeter reads zero.



The resistors and meters are reconnected to the supply, as shown in diagram 2.

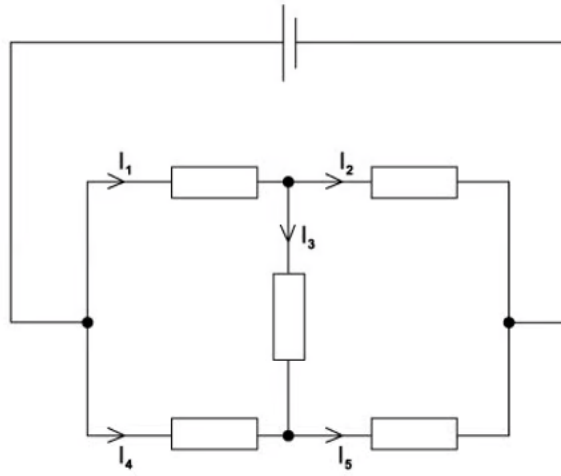
What are the meter readings in diagram 2?

	Voltmeter reading / V	Ammeter reading / A
A	0	1.0
B	3.0	0.5
C	3.0	1.0
D	6.0	0

[1 mark]

Question 9

The diagram shows currents I_1 , I_2 , I_3 , I_4 and I_5 in different branches of a circuit.



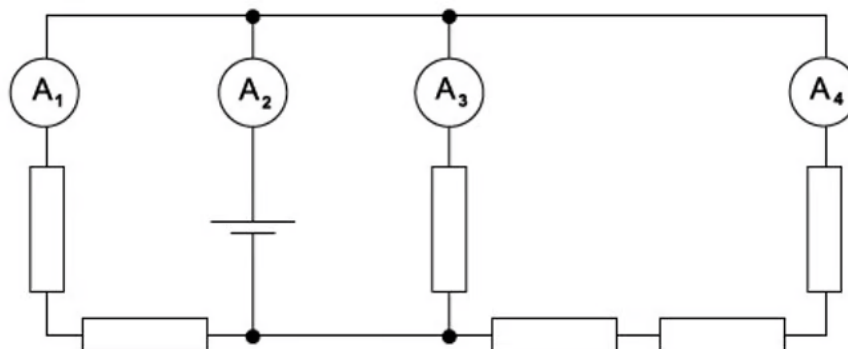
Which one of the following is correct?

- A. $I_1 = I_2 + I_3$
- B. $I_2 = I_1 + I_3$
- C. $I_3 = I_4 + I_5$
- D. $I_4 = I_5 + I_3$

[1 mark]

Question 10

In the circuit shown, all the resistors are identical and all the ammeters have negligible resistance.



The reading on ammeter A_1 is 0.6 A.

What are the readings on the other ammeters?

	Reading on ammeter A_2 / A	Reading on ammeter A_3 / A	Reading on ammeter A_4 / A
A	1.0	0.3	0.1
B	1.4	0.6	0.2
C	1.8	0.9	0.3
D	2.2	1.2	0.4

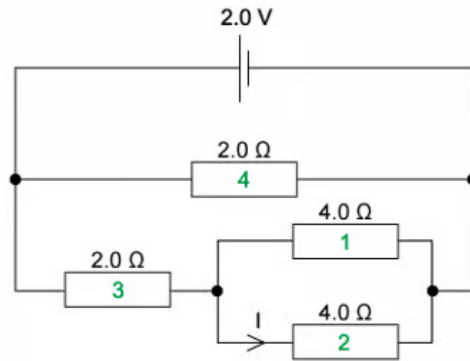
[1 mark]

Model Answers: Medium

1

The correct answer is **A** because:

- First, we must find the equivalent resistance of all the resistors combined
- We'll start with the two $4.0\ \Omega$ resistors, let's call these R_1 and R_2
- For resistors in parallel: $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$
 - $\frac{1}{R_{1+2}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$
 - So, $R_{1+2} = 2.0\ \Omega$
- Next, we'll add the combined R_1 and R_2 in series with R_3
- For resistors in series: $R = R_1 + R_2 + \dots + R_n$
 - So, $R_{1+2+3} = 2 + 2 = 4.0\ \Omega$
- And then, to get the total resistance, we'll add the combined 3 resistors in parallel with R_4
 - So, $\frac{1}{R_{total}} = \frac{1}{R_{1+2+3}} + \frac{1}{R_4} = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$
- Therefore, the total resistance is $R_{total} = \frac{4}{3}\ \Omega$
- Now, we must find the value of the current I through R_2
- Ohm's Law states $I = \frac{V}{R}$
 - So, the current through the cell is equal to $I = \frac{V}{R} = \frac{2.0\ V}{\frac{4}{3}\ \Omega} = 1.5\ A$
- If we consider there to be three resistors of $2.0\ \Omega$ each, the current through each resistor will be $\frac{2}{6} \times 1.5 = 0.5\ A$
- Since the resistance is equal in both branches, the current will divide equally
 - So, $I = \frac{0.5}{2} = 0.25\ A$



2

The correct answer is **D** because:

- The sum of p. d. across the internal and external resistors is equal to the e. m. f in the circuit
 - $E = I(R + r) = 12 \text{ V}$
 - So, the values must add up to 12 V
- Consider option D
 - Internal resistance $r = 2.0 \Omega$
 - External resistance $R = 4.0 \Omega$
 - Current $I = 1.8 \text{ A}$
- $E = I(R + r) = 1.8(4 + 2) = 10.8 \text{ V}$
- This is less than the stated e. m. f., so this combination is incorrect

A is incorrect because $E = I(R + r) = 1(10 + 2) = 12 \text{ V}$

B is incorrect because $E = I(R + r) = 1.2(8 + 2) = 12 \text{ V}$

C is incorrect because $E = I(R + r) = 1.5(6 + 2) = 12 \text{ V}$

3

The correct answer is **C** because:

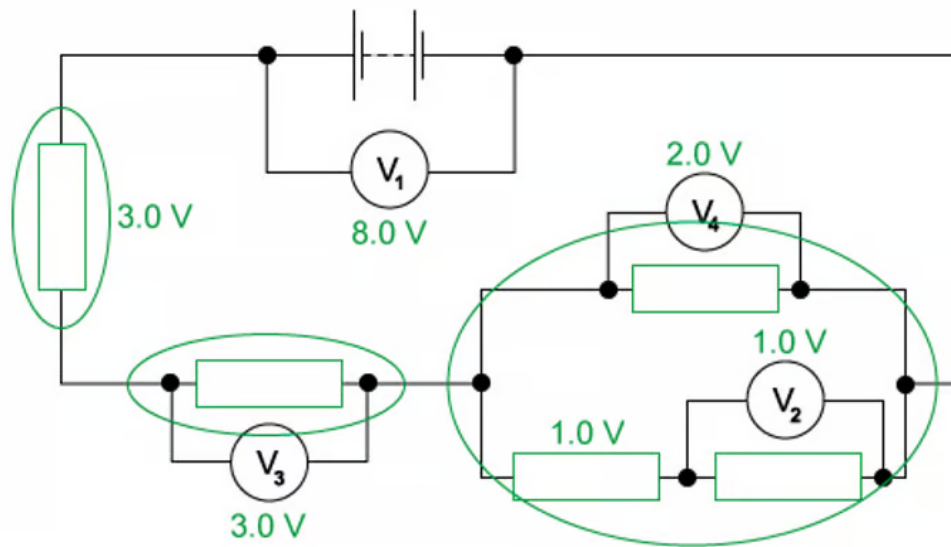
- Let the total resistance of the circuit be R
- For resistors in parallel: $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$

- $\frac{1}{R} = \frac{1}{1} + \frac{1}{2} + \frac{1}{5} = 1.7$
- So, the total resistance is $R = \frac{1}{1.7} = 0.588 \Omega$
- In other words, the three resistors in parallel is equivalent to a single resistor of resistance 0.588Ω
- A current of 5.0 A flows through this resistor
 - So, the p. d. across the branches $= IR = 5.0 \times 0.588 = 2.94 \text{ V}$
- As the resistors are in parallel, the p. d. across each branch is the same
 - $I = \frac{V}{R} = \frac{2.94}{2.0} = 1.47 \text{ A}$
- Therefore, the current through ammeter is 1.5 A

4

The correct answer is **B** because:

- Let the resistance of each resistor be R
- The p. d. across a resistor is $V_2 = 1.0 \text{ V}$
- The p. d. across the resistor just left of V_2 is also 1.0 V since the resistance is the same and they are in the same branch, so, the same current flows through them both
 - So, the total p. d. in the lower branch of the parallel combination is equal to $= 1.0 + 1.0 = 2.0 \text{ V}$
- The total p. d. across any branch of a parallel combination is the same according to Kirchhoff's Second Law
 - Therefore, $V_4 = 2.0 \text{ V}$
- The complete circuit may be described as a series combination of the two single resistor and the parallel combination (as shown in the diagram)
- The p. d. across the vertical resistor is the same as V_3 since they have the same resistance.
 - $V_1 = \text{total e. m. f in circuit} = 8.0 \text{ V}$
- The sum of p. d. in any loop is equal to the e. m. f in the circuit
 - $V_3 + V_3 + 2.0 = 8.0 \text{ V}$
 - $2V_3 = 6.0 \text{ V}$
- So, $V_3 = 3.0 \text{ V}$



5

The correct answer is **D** because:

- Ohm's Law states $V = IR$, so, the current in the whole circuit is $I = \frac{V}{R_T}$
 - Where R_T is the combined resistance of the resistors
- Let the resistance of a resistor = R and the e.m.f of the supply = E
- In circuit **D**, the 2 resistors are in a parallel combination, so the combined resistance is
 - $\frac{1}{R_T} = \frac{1}{R} + \frac{1}{R} = \frac{2}{R}$, therefore, $R_T = \frac{R}{2}$
- Therefore, the current in the circuit is $I = \frac{E}{\frac{R}{2}} = \frac{2E}{R}$
- This is the same current as circuit **C**, however, at a junction, the current splits
- In circuit **D**, the ammeter is connected directly to the positive terminal, so it reads the value of the current flowing in the whole circuit, whereas the current in **C** will read half the total current
 - Therefore, the ammeter in circuit **D** will have the greatest reading

A & B are incorrect because these two circuits are in a series arrangement, the current will be the lower than in a parallel arrangement: Circuit **A**: $R_T = R + R = 2R$, so,

the current $I = \frac{E}{2R}$

Reason for the correct answer:

- Therefore, the ammeter in circuit **D** will have the greatest reading

A & B are incorrect because these two circuits are in a series arrangement, the current will be the lower than in a parallel arrangement: Circuit **A**: $R_T = R + R = 2R$, so,

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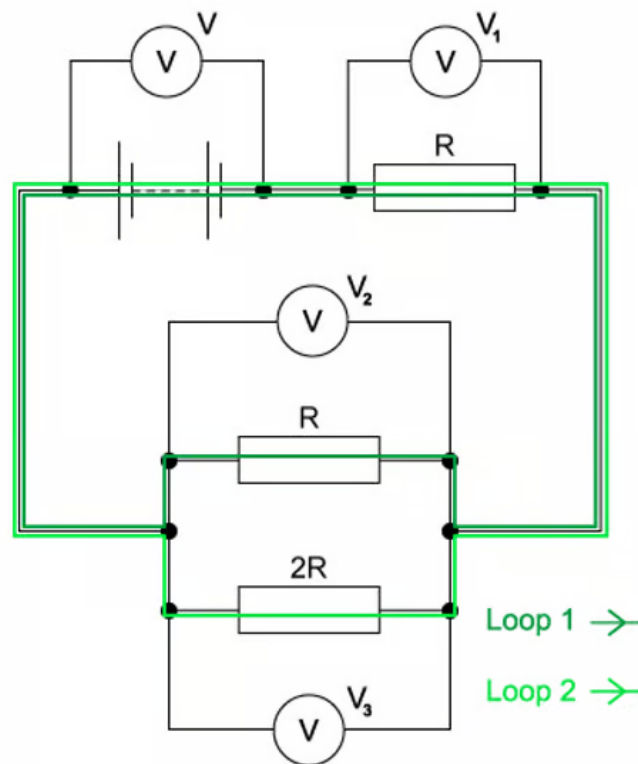
Circuit **B**: $R_T = R + 0 = R$, so, the current $I = \frac{E}{R}$

C is incorrect because in circuit **C** the current splits at a junction, so, the ammeter in circuit **C** gives only half the total current in the circuit

6

The correct answer is **D** because:

- The voltmeters are used to give the potential difference across the resistors. The voltmeter reading V gives the e.m.f of the battery
- The resistors connection in the circuit forms two different loops as shown in the diagram below (the voltmeters are not considered)
 - Loop 1: battery — resistor R (in parallel) — resistor R
 - Loop 2: battery — resistor $2R$ (in parallel) — resistor R
- From Kirchhoff's Laws, the sum of p. d. in any loop should be equal to the e. m. f in the circuit
 - $V = V_2 + V_1$
 - $V = V_3 + V_1$
 - $V - V_1 = V_3$



7

The correct answer is **C** because:

- Current is equal to $I = \frac{Q}{t}$
- A charge of 6.0 C flows through the resistor in a time of 2.0 minutes
- So, the current through the resistor is $I = \frac{6.0}{2 \times 60} = 0.05 \text{ A}$
- Power dissipated in a resistor is equal to $P = I^2 R$
- Power is also equal to $\frac{\text{energy}}{\text{time}}$
 - So, energy dissipated $= P \times t = (I^2 R) \times t = 48 \text{ J}$
 - $(0.05)^2 = R \times (2 \times 60) = 48$
 - Therefore, resistance $R = \frac{48}{(0.05)^2 \times (2 \times 60)} = 160 \Omega$
- The internal resistance of the battery and the resistor are in series, hence, the same current flows through them
- So, the sum of p. d. across these two resistors is equal to the e. m. f in the circuit
 - $E = I \times (R + r) = 9.0 \text{ V}$

- $(R + r) = \frac{9.0}{0.05} = 180 \, \Omega$
- $160 + r = 180$
- Therefore, the internal resistance is equal to $r = 180 - 160 = 20 \, \Omega$

8

The correct answer is **A** because:

- In diagram 1, the voltmeter reads zero because the potential difference across it is zero, meaning that both of its junctions are at the same potential
- From Kirchhoff's First Law, current splits at a junction
- Since the resistors are identical, the current splits equally into the two branches
 - The ammeter reads a current of 1.0 A, so the total current in the circuit is $1.0 + 1.0 = 2.0 \, \text{A}$
- Now, let's consider diagram 2: current flows from the positive terminal of the supply and is split equally at a junction, so, the ammeter still reads the current that has been split equally into two
 - This current is equal to $\frac{2.0}{2} = 1.0 \, \text{A}$
- From the junction connected to the positive terminal of the supply, the current splits at the bottom junction and rejoins again at the top junction to return to the negative terminal
- Since the same amount of current flows through identical resistors the potential of the left and right junctions would be the same
 - Therefore, the potential difference across the voltmeter is zero

B & D are incorrect because the total current in diagram 1 and 2 is 2.0 A so when the current splits at the junction, the reading on both ammeters is $\frac{2.0}{2} = 1.0 \, \text{A}$

C is incorrect because to detect a potential difference, the voltmeter would need to be connected across one component at a time

9

The correct answer is **A** because:

- Kirchhoff's First Law states that the total current entering a junction is equal to

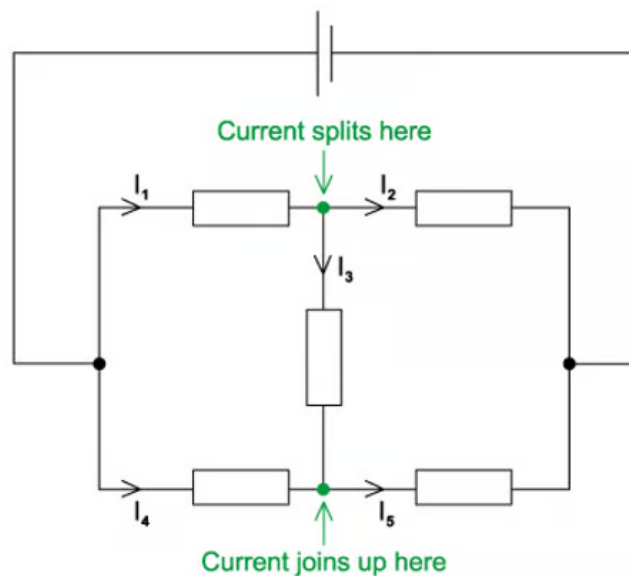
the total current leaving the same junction

- In other words the algebraic sum of all the current entering and leaving a junction must be equal to zero, so: $\sum I_{IN} = \sum I_{OUT}$
- At the top junction, indicated in the diagram, I_1 splits into I_2 and I_3

$$\circ I_1 = I_2 + I_3$$

- At the bottom junction, indicated in the diagram, I_3 and I_4 join up to become I_5

$$\circ I_5 = I_3 + I_4$$



B is incorrect because current I_2 is equal to $I_2 = I_1 - I_3$

C is incorrect because current I_3 is equal to $I_3 = I_5 - I_4$

D is incorrect because current I_4 is equal to $I_4 = I_5 - I_3$

10

The correct answer is **D** because:

- Current flows from the positive terminal of the battery, so, the current in A_2 should have the highest value
- Let the resistance of each resistor be R
- The left-hand side of the circuit, in the ($A_1 - A_2$) loop, there are two resistors in

D is incorrect because current I_4 is equal to $I_4 = I_5 - I_3$

10

The correct answer is **D** because:

- Current flows from the positive terminal of the battery, so, the current in A_2 should have the highest value
- Let the resistance of each resistor be R
- The left-hand side of the circuit, in the ($A_1 - A_2$) loop, there are two resistors in series, so their total resistance is $R + R = 2R$
 - The current in the ($A_1 - A_2$) loop = $0.6A$, as read by ammeter A_1
- So, from Ohm's Law $V = IR$, the potential difference in the ($A_1 - A_2$) loop is equal to $V = 0.6(2R) = 1.2R$
- From Kirchhoff's Second Law, the potential difference is the same as the e.m.f. in the circuit for any loop, so, the e.m.f. in the circuit is $1.2R$, and this is also the potential difference across all loops
- Consider the ($A_2 - A_3$) loop:
 - Current in $A_3 = \frac{V}{R} = \frac{1.2R}{R} = 1.2A$
- Similarly, consider the ($A_2 - A_4$) loop:
 - Current in $A_4 = \frac{1.2R}{3R} = 0.4A$
- Therefore, current in $A_2 = \text{total current} = 0.6 + 1.2 + 0.4 = 2.2A$

